

Operand Extension and Address Extension

Operand Extension

Most CPUs operate on fixed-size data words (e.g., 32-bit or 64-bit). However, instruction operands may be shorter (e.g., 8-bit or 16-bit), especially in RISC architectures. These shorter operands need to be extended to match the CPU word size before operations can be performed. There are two primary techniques:

- **Sign Extension**
- **Zero Extension**

Sign Extension

Used for **signed numbers** (two's complement). It preserves the sign and magnitude by replicating the sign bit (MSB).

Procedure:

- Let N be an m -bit signed number.
- Let $n > m$ be the CPU word size.
- Replicate the most significant bit (MSB), also the sign bit s , $(n - m)$ times on the left.

Example:

$$\begin{aligned} N &= 10101 \ 01010101 \quad (13\text{-bit}) \\ \text{Sign-Extended } N &= \underbrace{11111111111111111}_{19 \text{ bits}} 010101010101 \\ N &= -2731_{10} \end{aligned}$$

Zero Extension

Used for **unsigned numbers**. The value is extended by adding leading zeros.

Procedure:

- Let N be an m -bit unsigned number.
- Extend to n bits by prefixing $(n - m)$ zeros.

Example:

$$N = 10101 \ 01010101 \quad (13\text{-bit})$$

$$\text{Zero-Extended } N = \underbrace{00000000000000000000}_{19 \text{ bits}} 010101010101$$

Address Extension

Some instruction formats have short address fields due to fixed instruction length (e.g., 13-bit offset), which is smaller than the actual addressable memory space (e.g., 32-bit).

Problem: Limited Range

Zero-extending a short address directly allows access to only 2^m locations.

Solution: Base + Offset Addressing

- Treat the m -bit address as an **offset**.
- Keep a full-size **base address** in a register.
- Compute effective address as:

$$\text{Effective Address} = \text{Base Register} + \text{Zero-Extended Offset}$$

Instruction Example

$$\text{STB } R_s, R_d(S_2) \Rightarrow M[R_d + S_2] := R_s[24:31]$$

Explanation:

- S_2 is a short (e.g., 13-bit) offset.
- R_d holds the base address (32-bit).
- $R_s[24:31]$ is the byte to store.
- Effective address = $R_d + \text{Zero-Extended}(S_2)$

Summary Table

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Extension Type	Used For	Method	Example
Sign Extension	Signed Numbers	Replicate sign bit to the left	1010101010101 → 111111...010101010101
Zero Extension	Unsigned Numbers	Add leading zeros	1010101010101 → 000000...010101010101
Address Extension	Short Address Fields	Zero-extend offset, add base register	Effective Address = Base + Offset